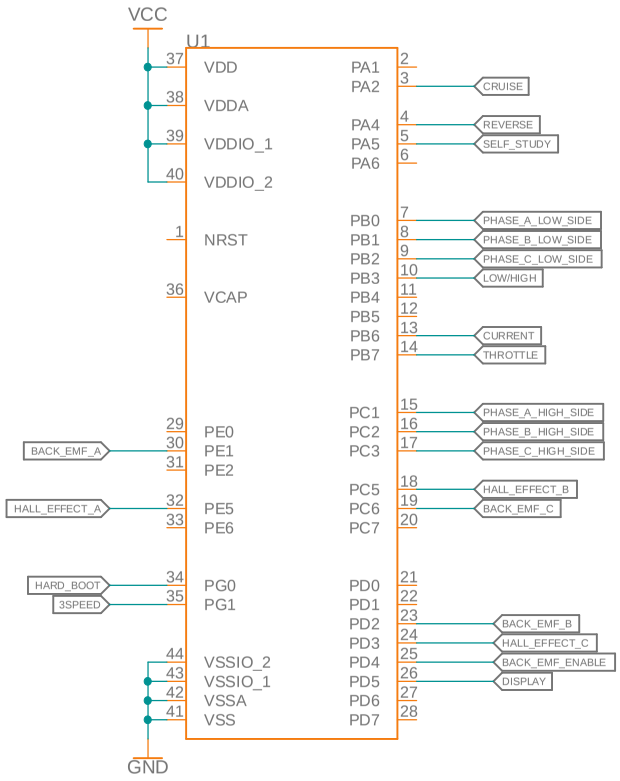
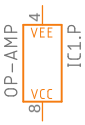
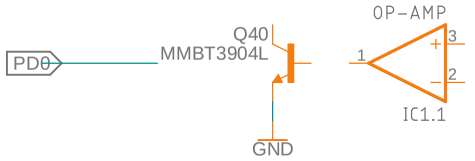




Not sure what's going on

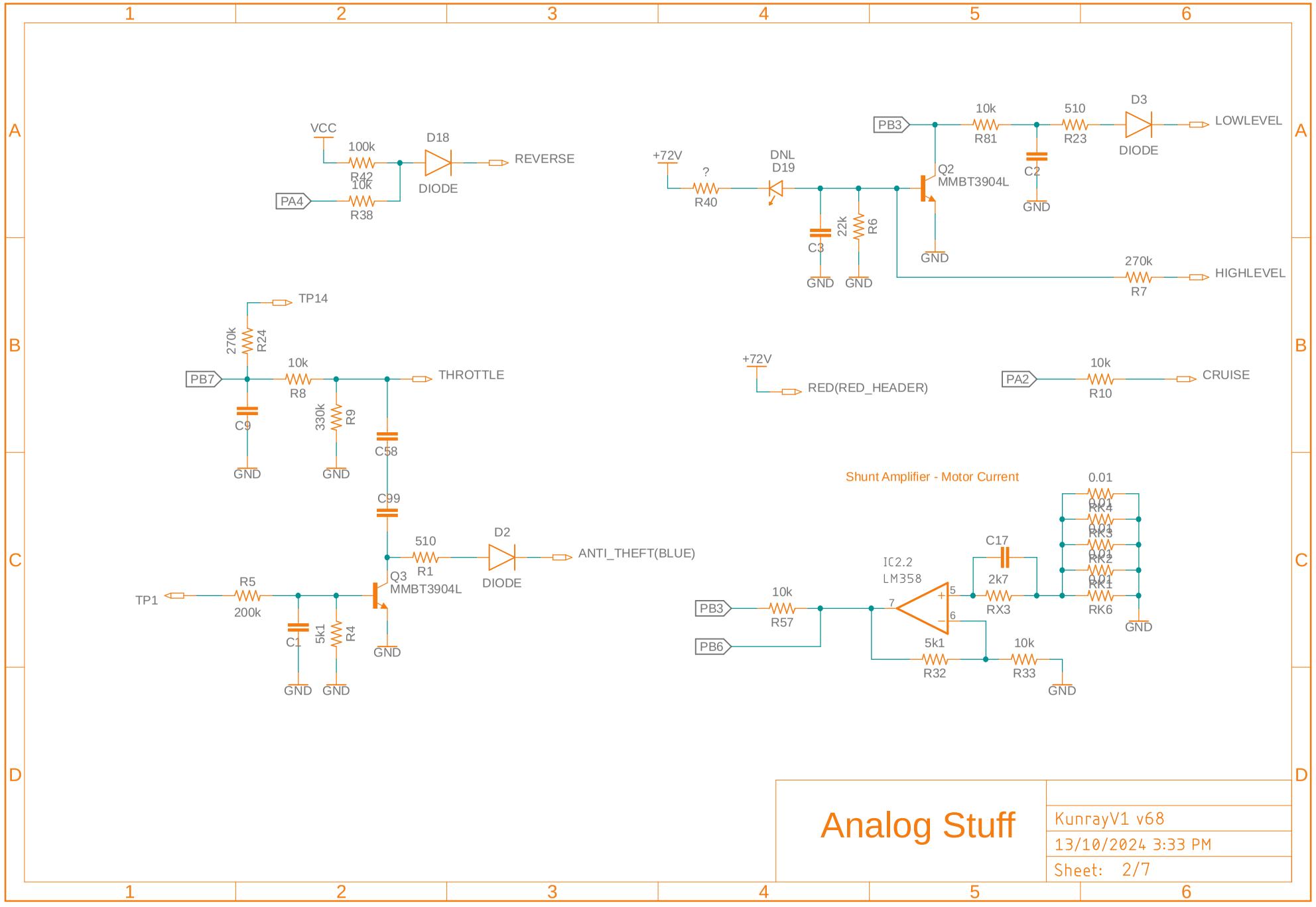


Processor

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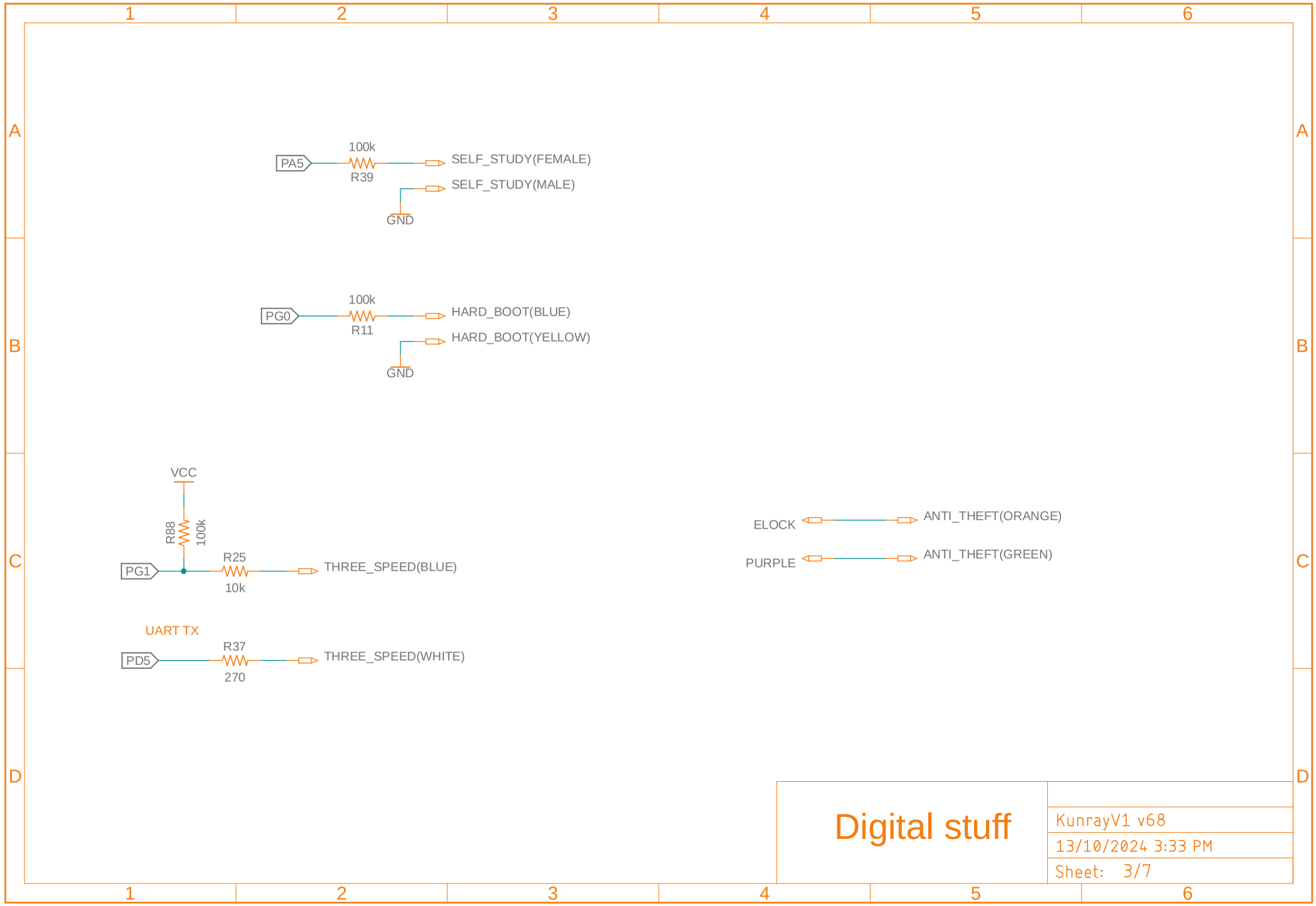


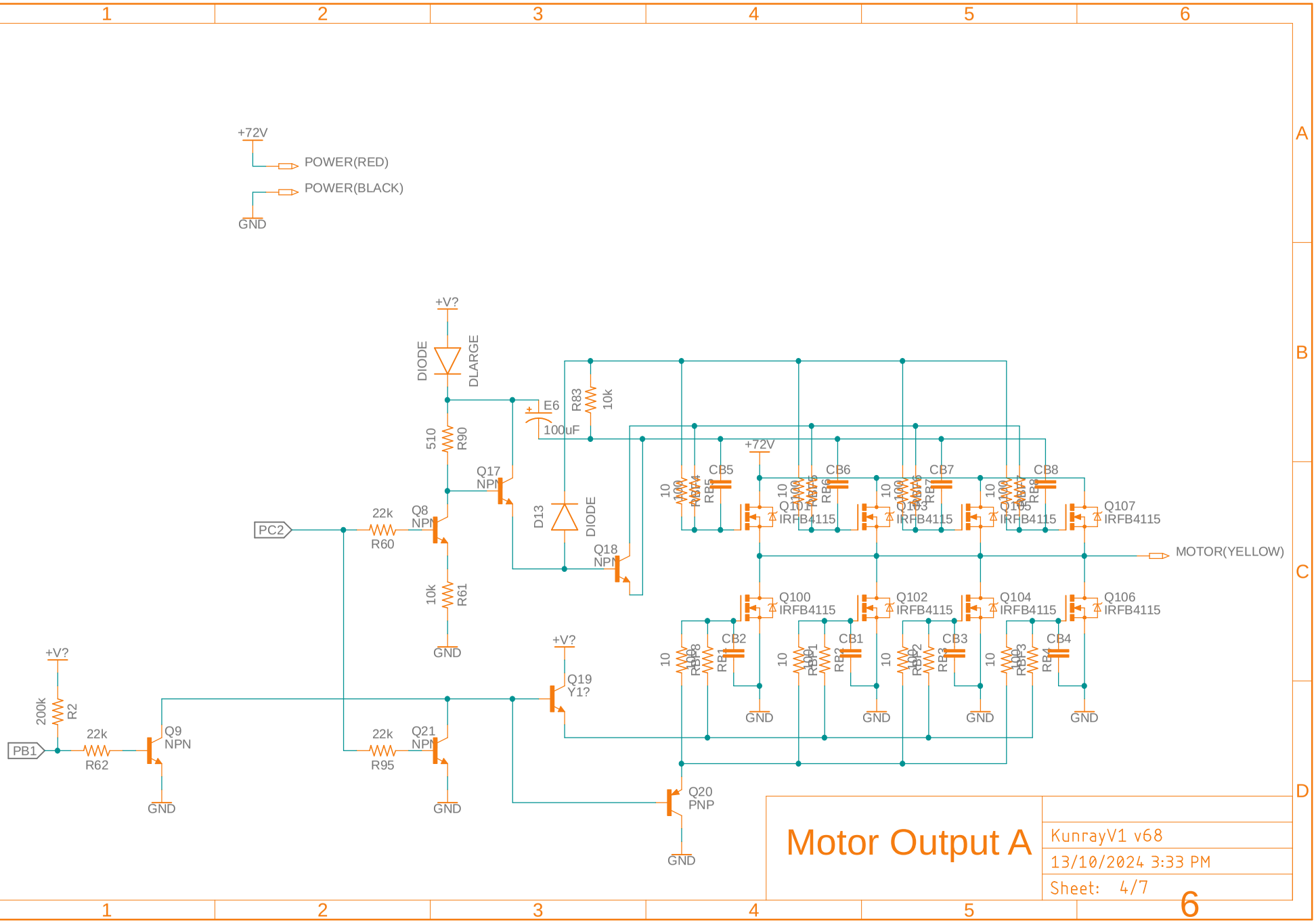
Analog Stuff

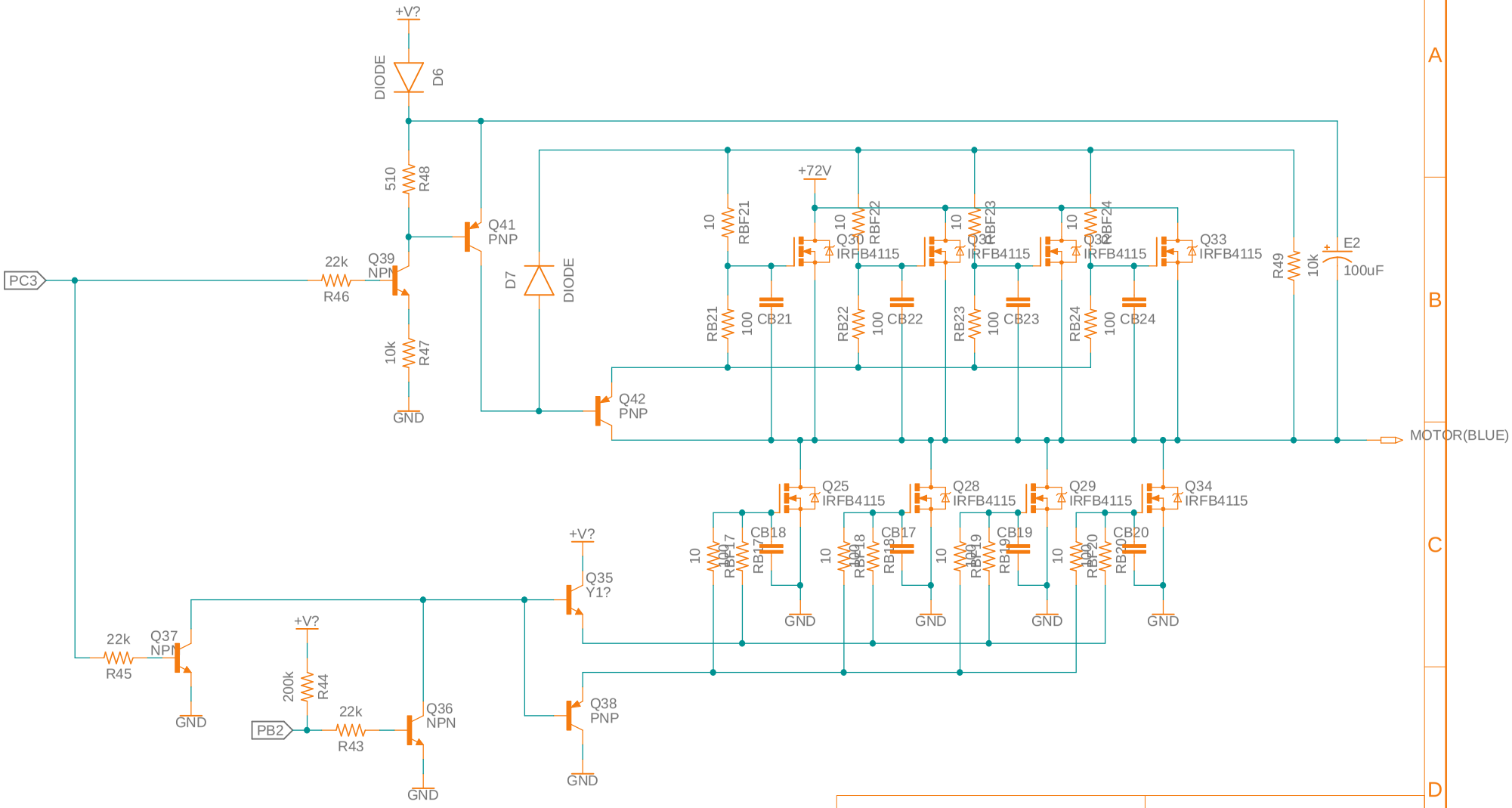
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1
2
3
4
5
6
A
B
C
D
PA5
100k
R39
SELF_STUDY(FEMALE)
SELF_STUDY(MALE)
GND
PG0
100k
R11
HARD_BOOT(BLUE)
HARD_BOOT(YELLOW)
GND
VCC
R88
100k
PG1
R25
10k
THREE_SPEED(BLUE)
UART TX
PD5
R37
270
THREE_SPEED(WHITE)
ELOCK
ANTI_THEFT(ORANGE)
PURPLE
ANTI_THEFT(GREEN)
Digital stuff
KunrayV1 v68
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+72V
POWER(RED)
POWER(BLACK)
GND
+V?
DIODE
DLARGE
E6
100uF
R83
10k
510
R90
Q17
NPN
D13
DIODE
Q18
NPN
PC2
22k
R60
Q8
NPN
10k
R61
GND
+72V
CB5
CB6
CB7
CB8
Q101
IRFB4115
Q103
IRFB4115
Q105
IRFB4115
Q107
IRFB4115
MOTOR(YELLOW)
Q100
IRFB4115
Q102
IRFB4115
Q104
IRFB4115
Q106
IRFB4115
CB2
CB1
CB3
CB4
+V?
Q19
Y1?
200k
R2
PB1
22k
R62
Q9
NPN
22k
R95
Q21
NPN
Q20
PNP
Motor Output A
KunrayV1 v68
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6

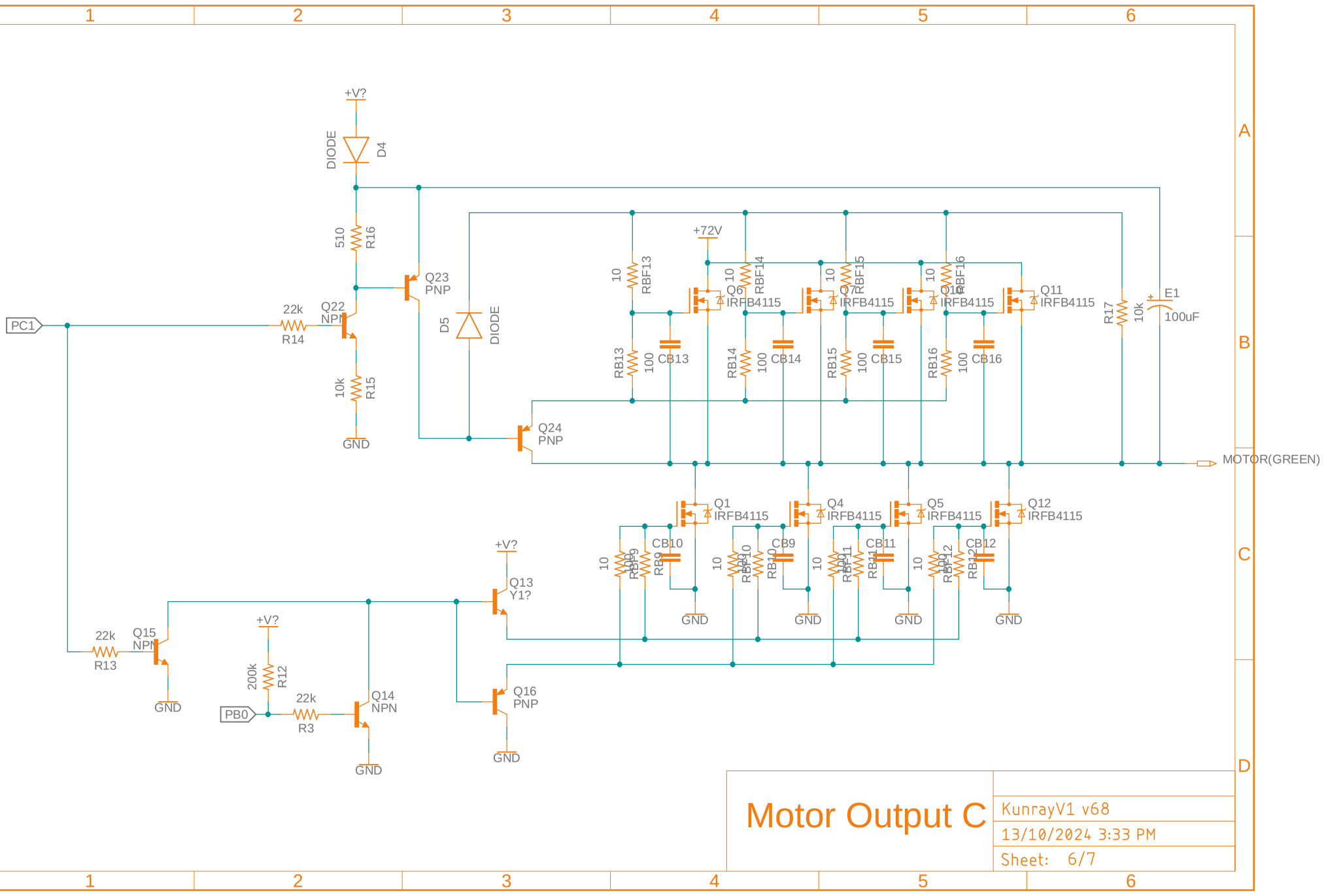


Motor Output B

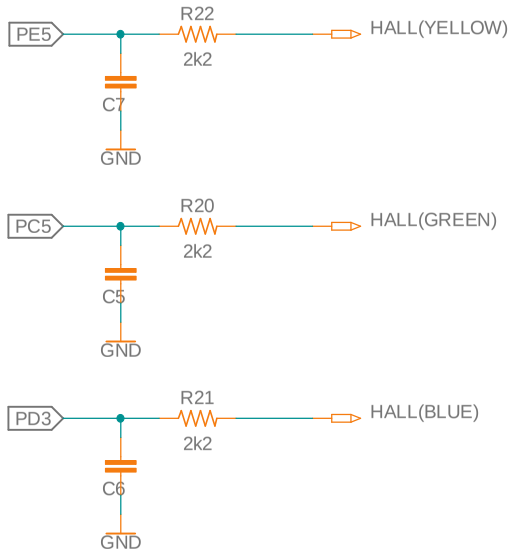
KunrayV1 v68

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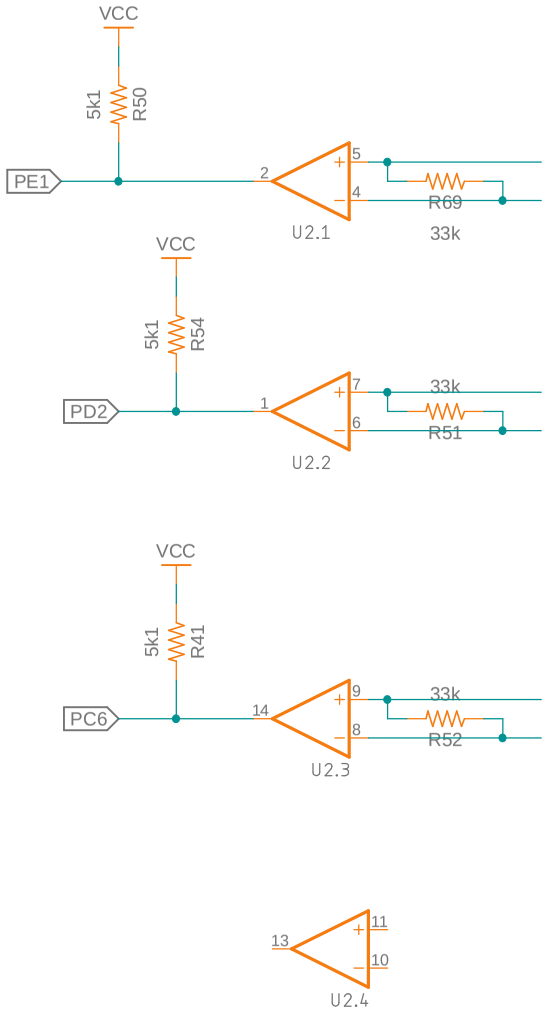
Sheet: 5/7


+V?
DIODE
D4
510
R16
Q23
PNP
PC1
22k
R14
Q22
NPN
10k
R15
GND
D5
DIODE
+72V
10
RBF13
RB13
100
CB13
Q6
IRFB4115
RBF14
RB14
CB14
Q7
RBF15
RB15
CB15
Q10
RBF16
RB16
CB16
Q11
IRFB4115
R17
10k
E1
100uF
MOTOR(GREEN)
Q24
PNP
Q1
IRFB4115
Q4
Q5
Q12
CB10
CB9
CB11
CB12
+V?
Q13
Y1?
Q16
PNP
22k
R13
Q15
NPN
200k
R12
PB0
22k
R3
Q14
NPN
Motor Output C
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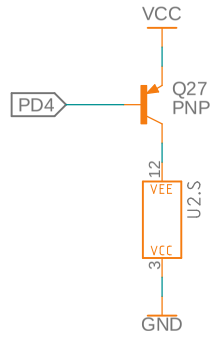
J1 Header - Hall Effect Sensors



Back EMF Sensing



not exactly, but you get the idea



Commutation

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